

Drone Challenge (UAV) Carnival Games Challenge

Ewing, NJ

Team #2255-1

TSA State Conference

TCNJ, NJ



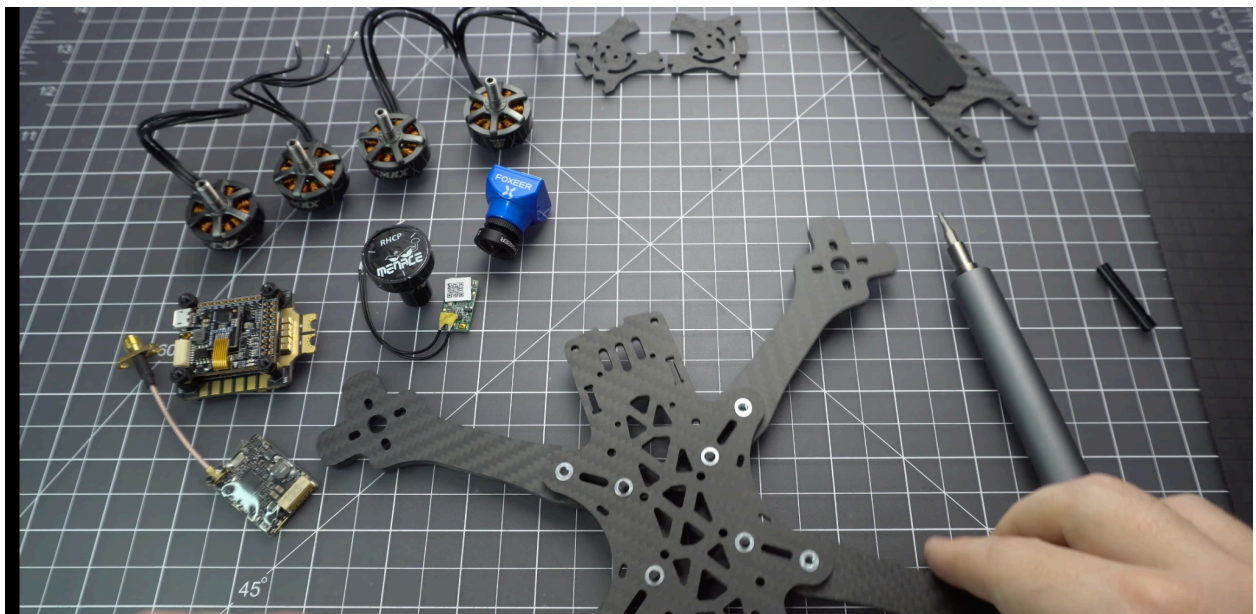
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Photo Log

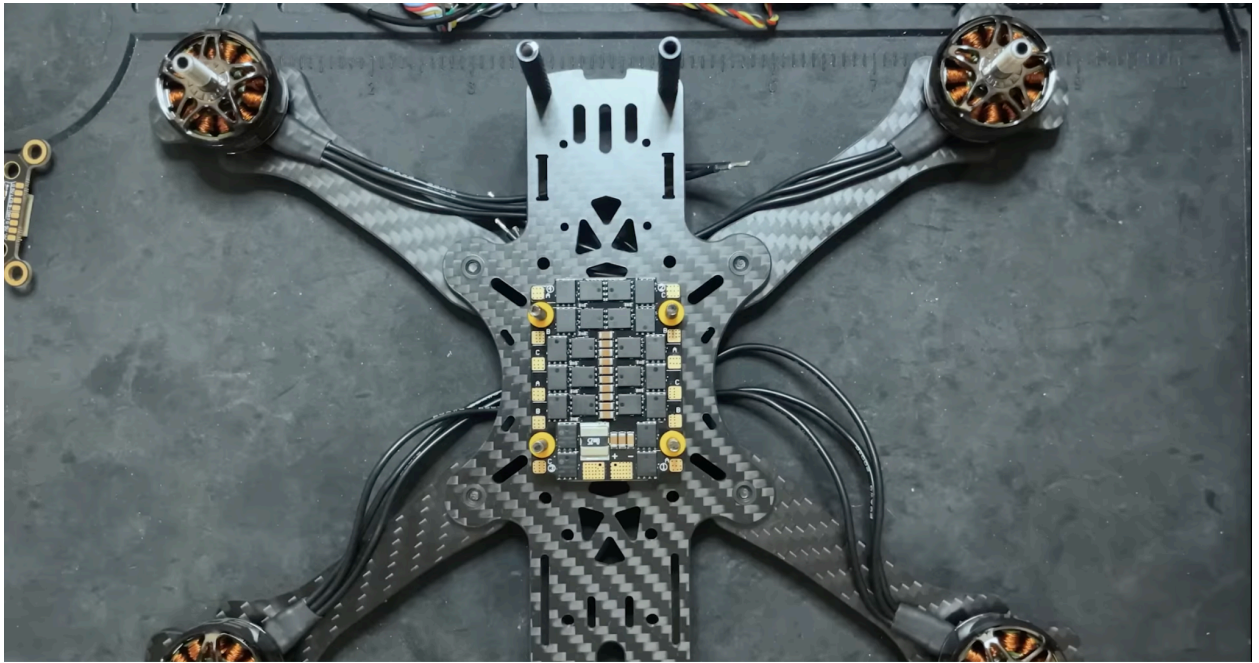
The assembly process of our FPV drone for the UAV Drone Carnival Games project involved several steps, starting from gathering the components to integrating the systems and testing the drone. Below is a detailed explanation of the assembly process:

1. **Frame Assembly:** We began by assembling the TBS Source One V5 5" frame, attaching the arms to the bottom plate, and securing them with screws. We then installed the standoffs and mounted the top plate to create a solid base for the rest of the components.



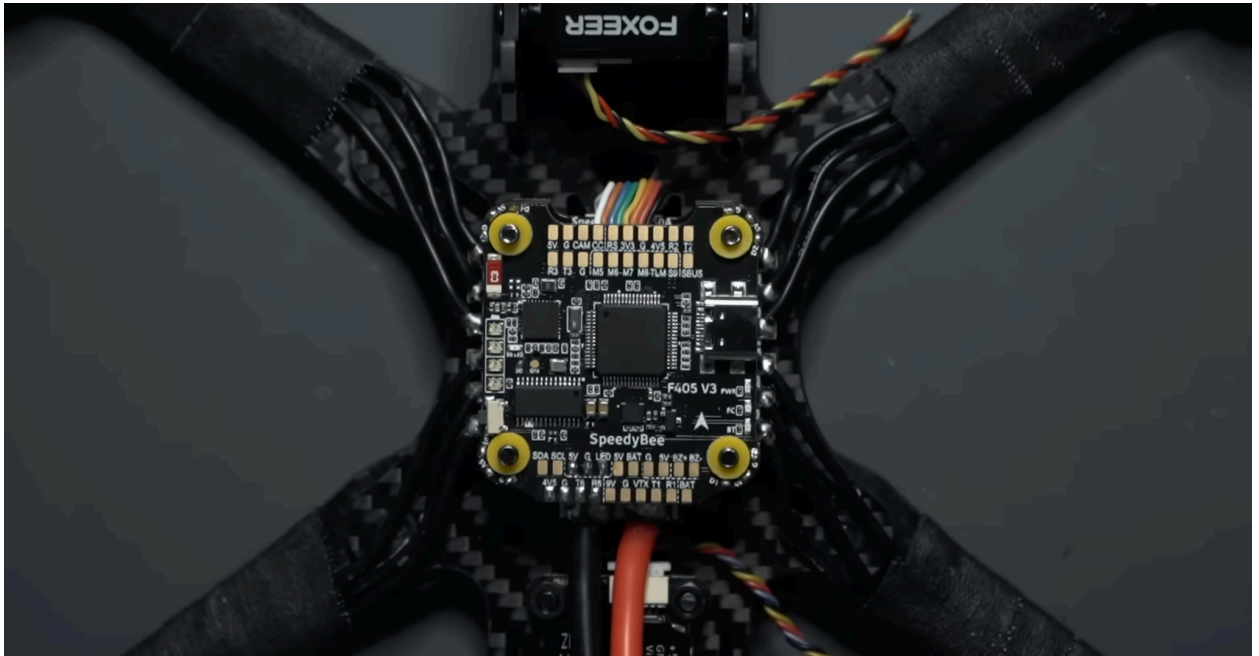
Components of the Drone

2. **Motors and ESC Installation:** Next, we mounted the iFlight XING-E Pro 2207 motors onto the drone's arms, securing them with screws. We then installed the iFlight BLITZ Stack - E55 4-in-1 55A ESC onto the frame, using nylon standoffs for insulation and vibration damping. We soldered the motor wires to the corresponding pads on the ESC.



Frame and ESCs of the Drone

3. Flight Controller Installation: We mounted the iFlight BLITZ Stack F7 V1.1 flight controller on top of the ESC, aligning the pin headers or connecting the two components using the provided cable. We secured the flight controller with nylon standoffs, ensuring a stable and vibration-free installation.
4. Receiver and GPS Module Integration: We connected the Happymodel TCXO ExpressLRS 2.4GHz Nano Receiver to the flight controller, soldering the appropriate wires to the designated UART pads. We also installed the HGLRC M100 Mini GPS module onto the frame, connecting it to the flight controller using the provided cable.



Flight Controller of the Drone

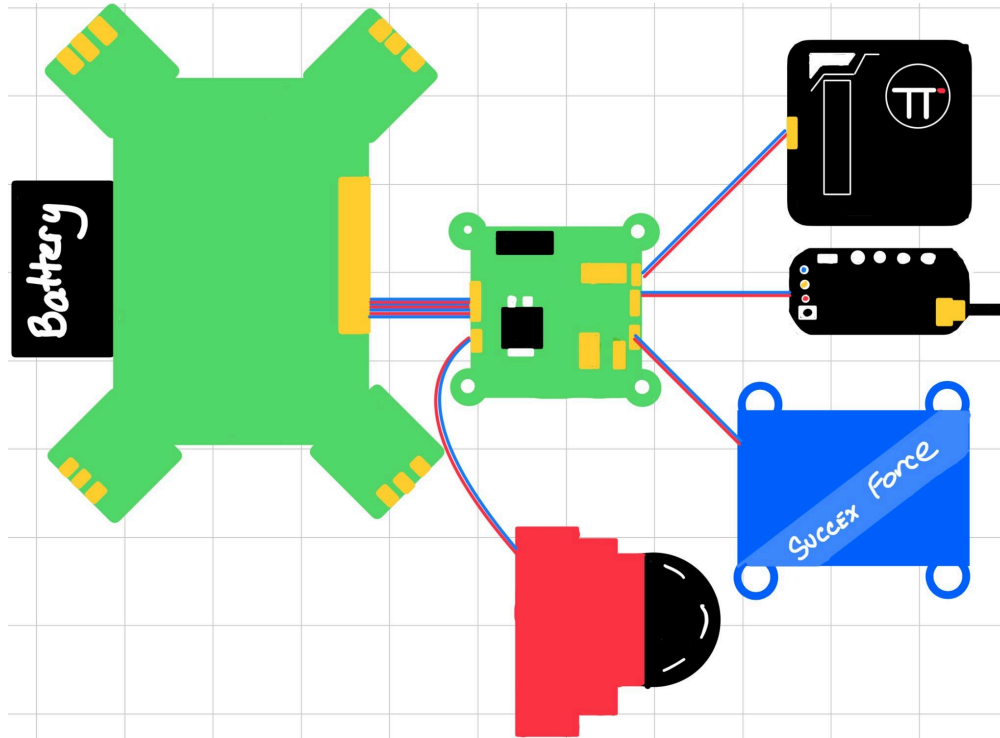
5. Video System Setup: We mounted the Caddx Ratel 2 Micro Starlight 1200TVL FPV camera to the front of the frame using the provided camera mount. We then installed the FPV VTX SpeedyBee TX800 onto the frame and connected it to the flight controller. We soldered the video signal, power, and ground wires to their respective pads.



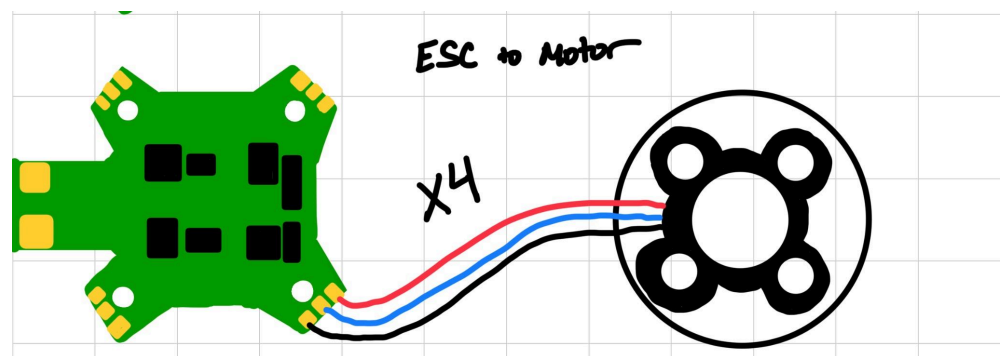
Camera of the Drone

6. **Power System Setup:** We installed the CNHL Black Series 1300mAh 6S Lipo Battery 22.2V 100C onto the top plate of the frame, using a battery strap to secure it in place. We connected the battery to the ESC via an XT60 connector, ensuring a solid connection for power delivery.
7. **Propellers and Grabber Mechanism:** We attached the iFlight Nazgul F5 5x3.5x3 3-Blade 5.1" Propellers to the motors, making sure to match the propeller's rotation direction with the motor's rotation. We also installed the grabber mechanism on the bottom of the drone, powered by a servo, and connected it to a servo controller that is powered by the flight controller.
8. **Final Checks and Configurations:** After assembling the drone, we double-checked all connections, ensuring that the wiring was neat and secure. We then connected the drone to a computer and used Betaflight Configurator to program the flight controller, set up the receiver, adjust PID values, and configure the various modes and auxiliary functions.

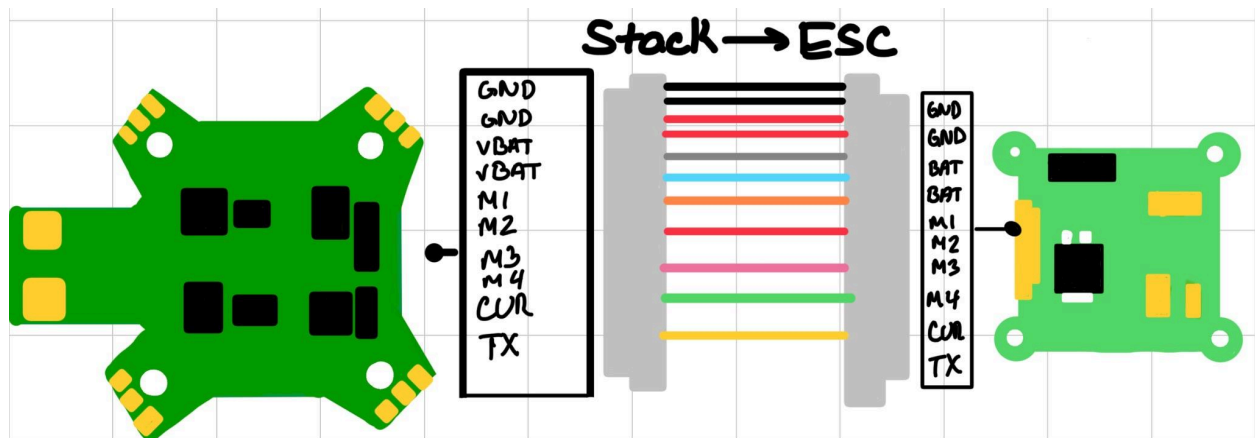
Wiring Diagrams



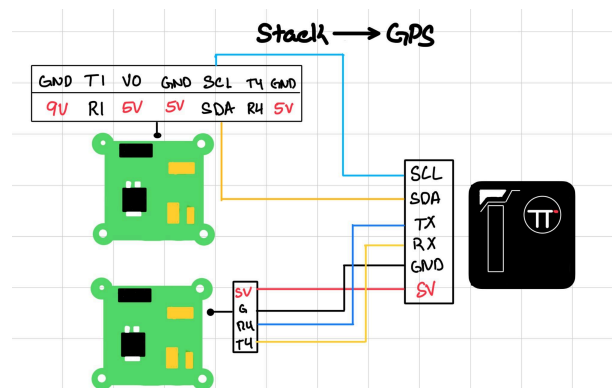
Full Diagram



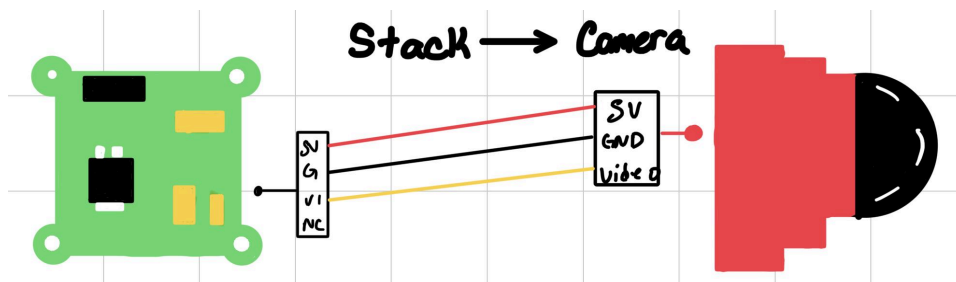
ESC to Motor Connection (x4)



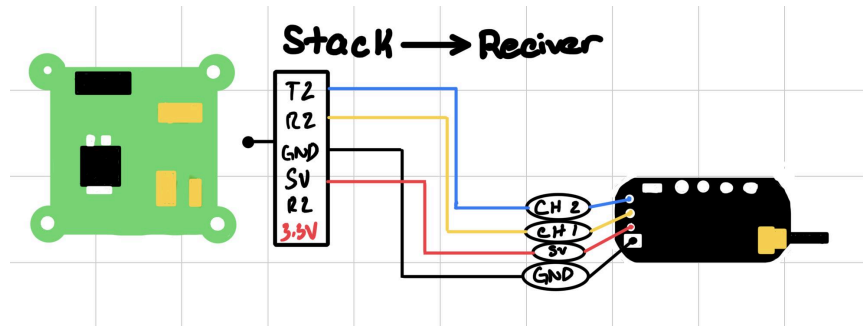
Flight Controller to ESC



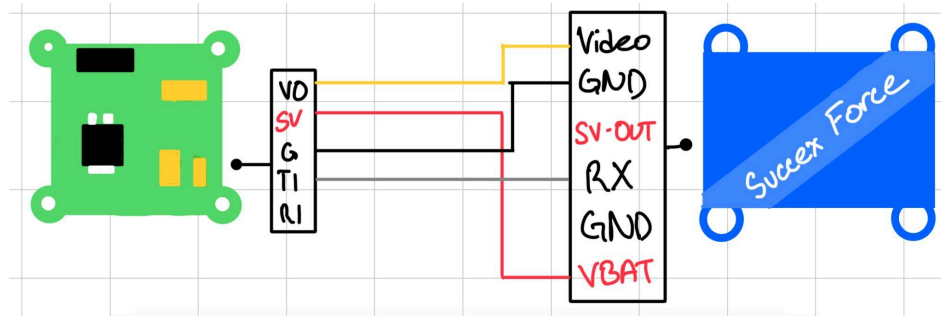
Flight Controller to GPS



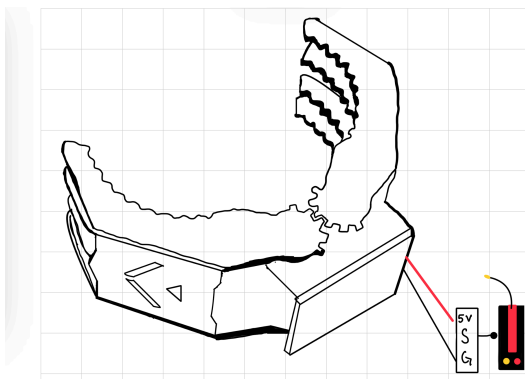
Flight Controller to Camera



Flight Controller to Receiver



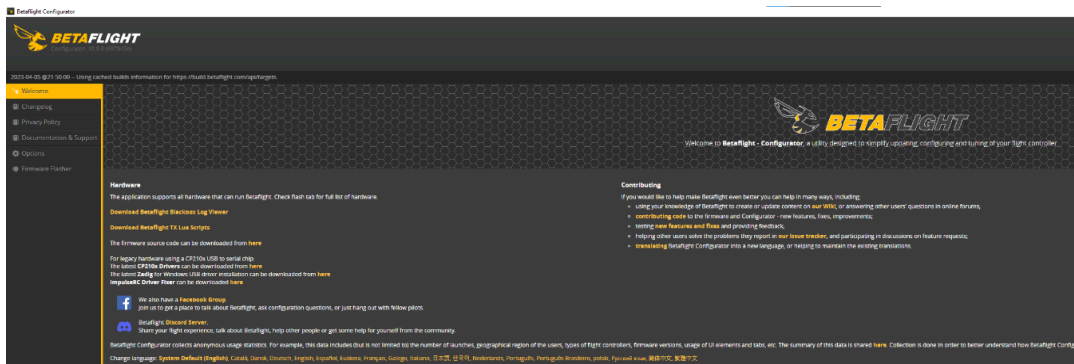
Flight Controller to VTX



← *Claw to Servo Transmitter*

Explanation of Programming

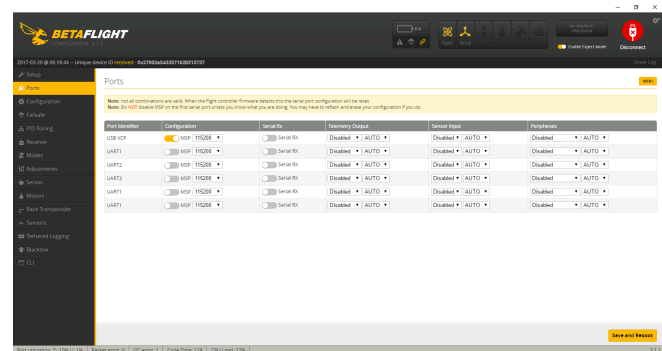
Betaflight is an open-source flight controller software designed for FPV drones. It provides an intuitive interface for users to configure, adjust, and fine-tune the performance of their drones. Our team utilized Betaflight to program our drone, ensuring optimal flight characteristics and ease of control. The following sections explain the steps we took and the various settings we adjusted using Betaflight.



Betaflight Configurator

During the initial setup, after assembling the drone, we connected the flight controller to a computer via USB. We then launched the Betaflight Configurator, which automatically detected our flight controller and established a connection. We verified the drone's orientation and calibrated the accelerometer to ensure accurate measurements of the drone's position and movements.

In the Ports tab, we configured the UART connections for the GPS module, receiver, and video transmitter. We enabled the Serial RX for the corresponding UART where our receiver was connected and set the correct protocol for GPS and VTX to ensure proper communication between the components.

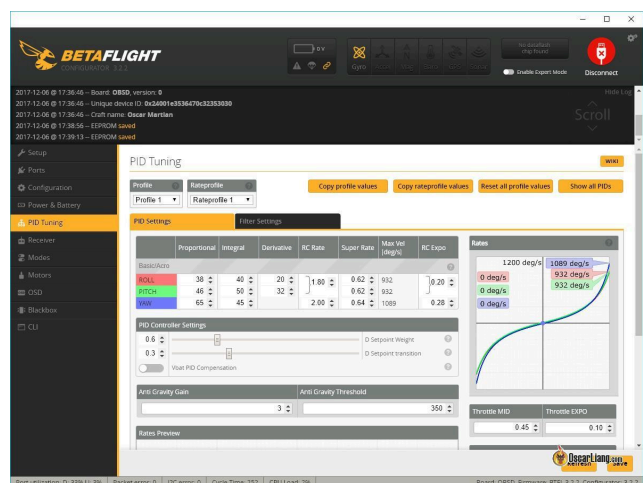


Ports Tab

We visited the Configuration tab to set up essential features such as the motor protocol (e.g., DSHOT), arming angle, and system configuration (e.g., gyro update frequency). Additionally, we enabled GPS, telemetry, and the appropriate receiver protocol (e.g., ExpressLRS). We also adjusted the failsafe settings to ensure a safe and controlled response in case of signal loss.

The PID Tuning tab allowed us to adjust the drone's flight characteristics. By fine-tuning the proportional, integral, and derivative (PID) values, we enhanced the drone's stability and responsiveness, making it easier to control during the carnival games. We performed several test flights and iteratively adjusted the PIDs to achieve the desired flight performance.

In the Receiver tab, we verified that our transmitter's input was correctly mapped to the corresponding channels, ensuring proper control over the drone's movements. We also set up the channel mapping to match our Radiomaster Zorro transmitter's configuration.



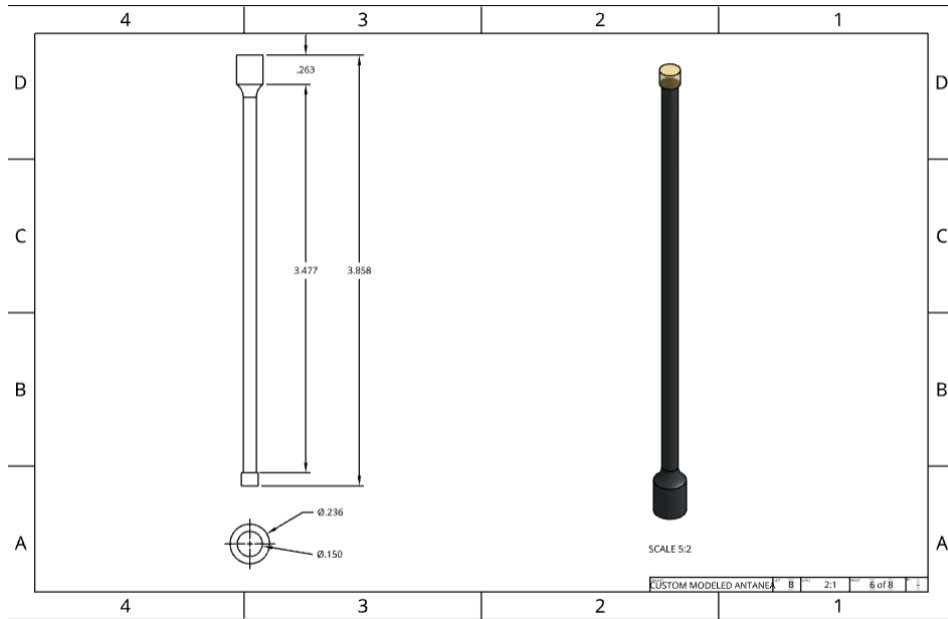
PID Tuning Tab

We used the Modes tab to assign various flight modes and auxiliary functions to specific switches on our transmitter. By setting up modes such as Angle, Horizon, and Acro, we enabled different levels of flight assistance, making it easier for the drone to navigate through the carnival games. Additionally, we configured switches for arming/disarming the drone, enabling GPS Rescue, and controlling the grabber mechanism.

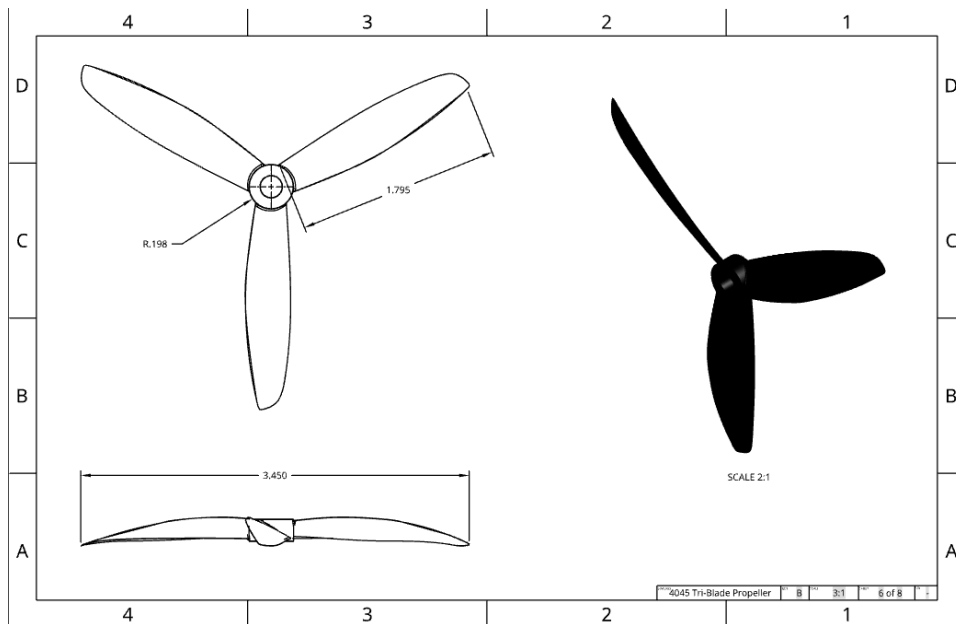
In the OSD (On-Screen Display) tab, we customized the information displayed on our FPV goggles, such as battery voltage, flight time, and GPS coordinates. This real-time data helped the pilot monitor the drone's status and make informed decisions during the carnival games.

By leveraging Betaflight's powerful features and user-friendly interface, our team successfully programmed the drone and optimized its performance for the UAV Drone Carnival Games project. The software's various controls allowed us to fine-tune the drone's flight characteristics, making it easier for the pilot to navigate through the challenging game scenarios.

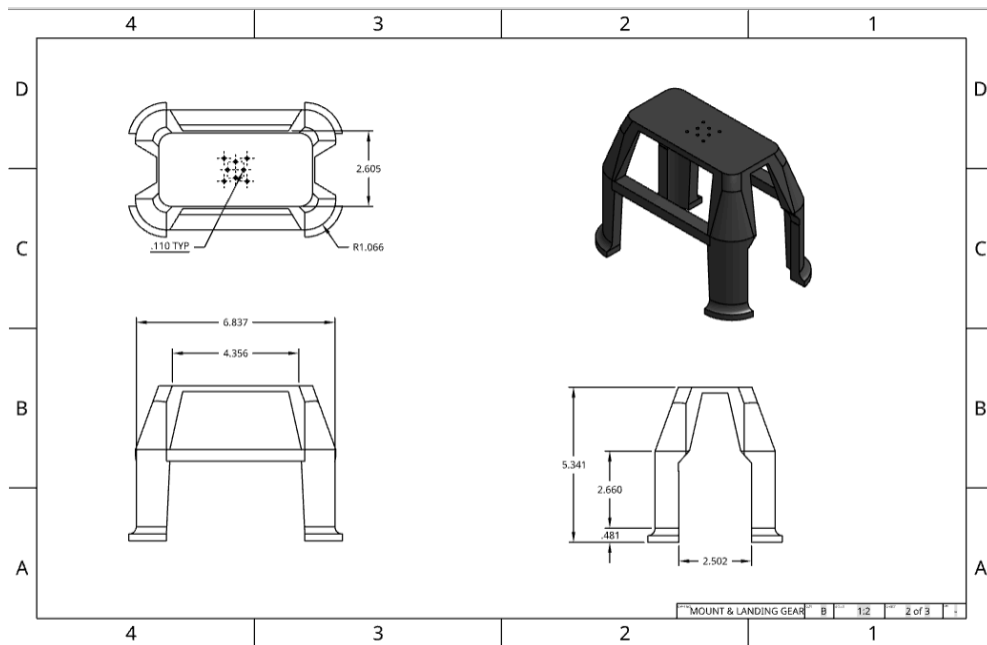
Engineering Drawings



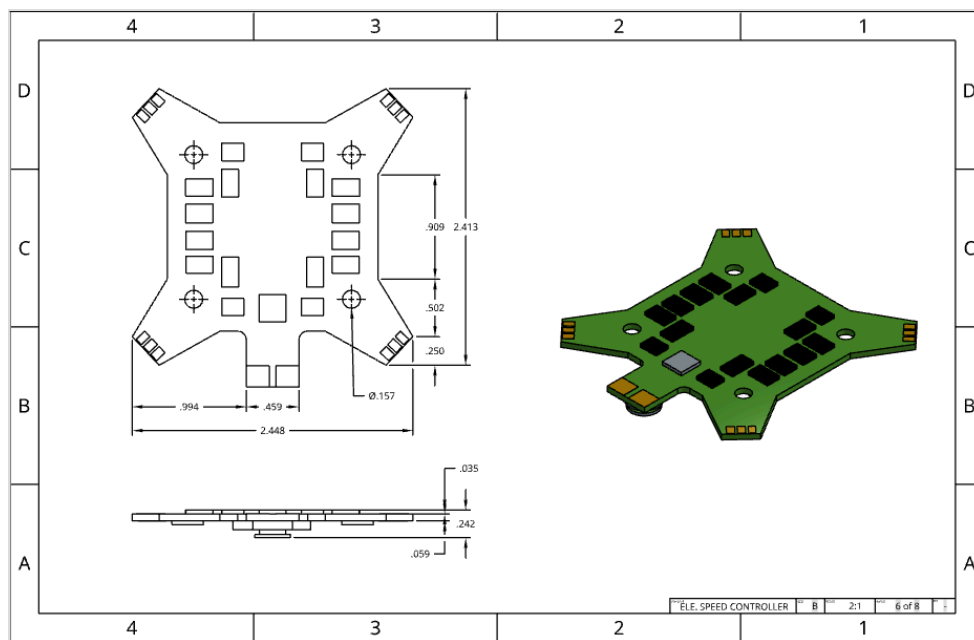
VTX Antenna



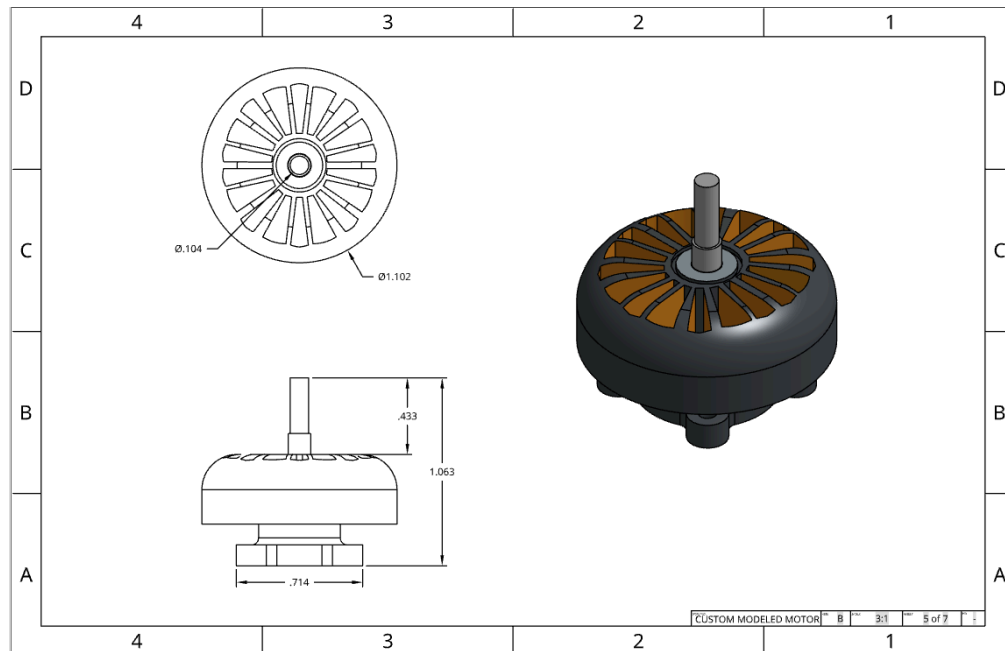
5- Inch Propeller



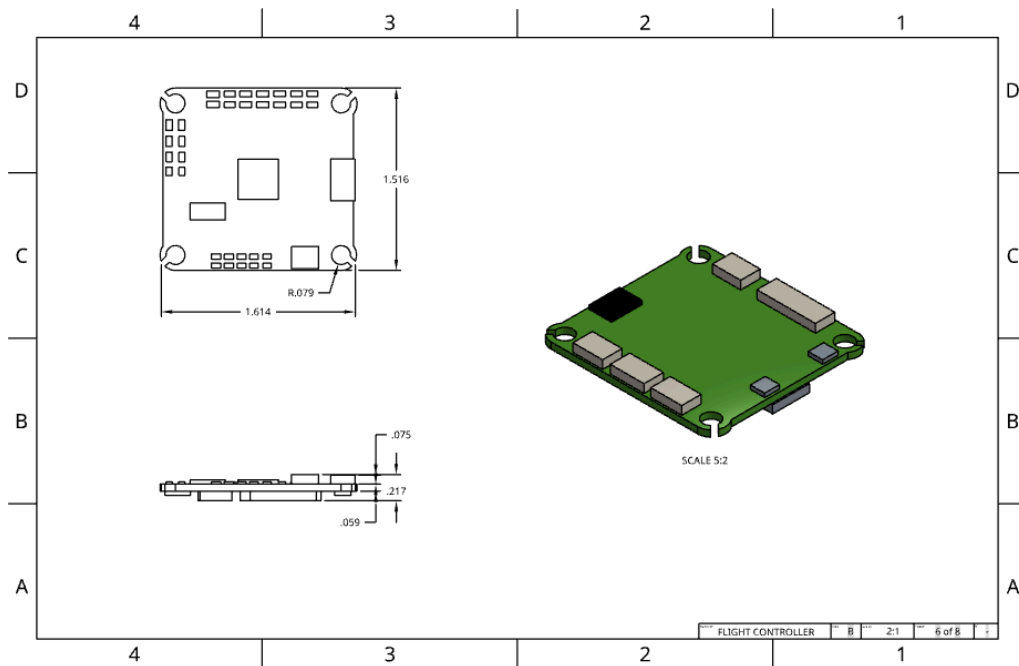
Mount & Landing Gear



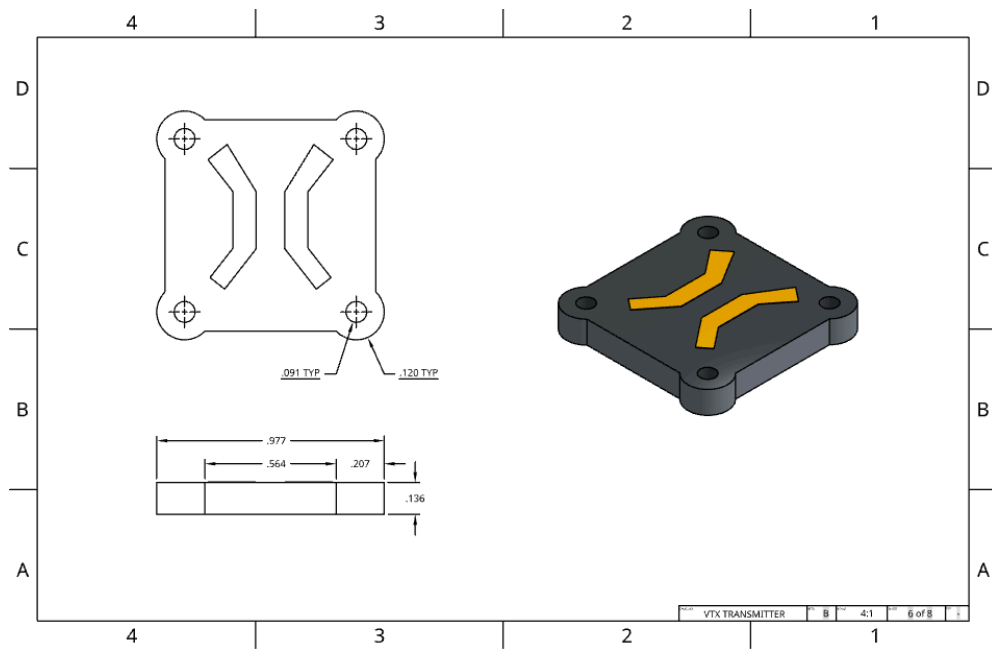
Electronic Speed Controller (ESC)



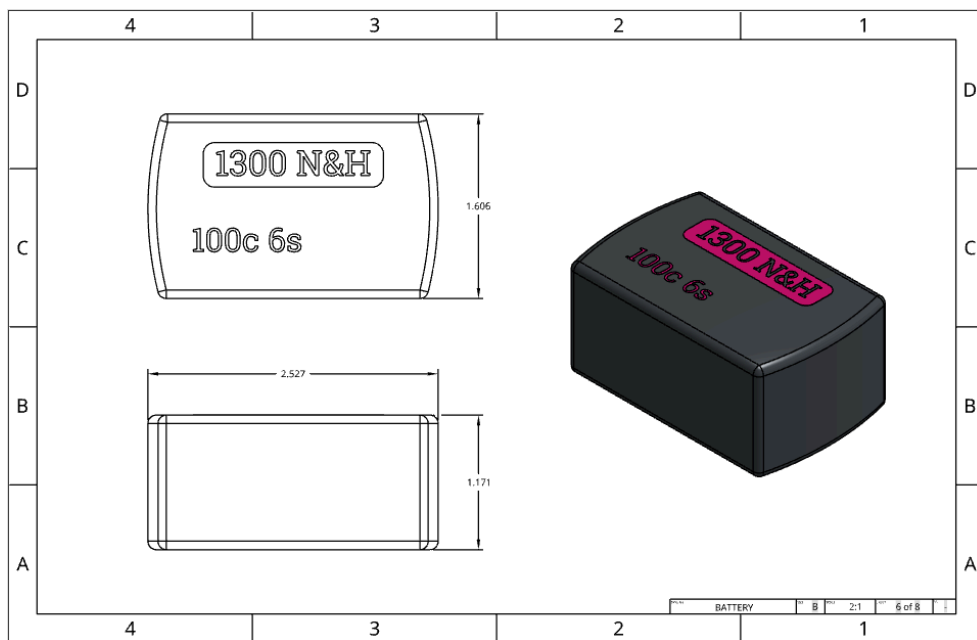
Brushless Motor



Flight Controller (FC)



VTX Transmitter



Battery

Documentation of Parts & Bill of Materials

General Part	Specific Part Named	Unit Price	Quantity	Total Amount
Frame	TBS Source One V5 5"	\$62.57	2	\$125.14
AIO Stack (FC and ESC)	iFlight BLITZ Stack - F7 V1.1 FC + E55 4-in-1 55A ESC	\$155.99	1	\$155.99
Motors	iFlight XING-E Pro 2207 Motor (1800kv)	\$62.99	2	\$125.98
Propellers	iFlight Nazgul F5 5x3.5x3 3-Blade 5.1" Propeller (Set of 4)	\$13.99	3	\$41.97
Battery	CNHL Black Series 1300mAh 6S Lipo Battery 22.2V 100C	\$63.99	3	\$191.97
Battery Charger	Gens Ace Imars III PRO 100W 10A AC Smart Balance Battery Charger	\$59.99	1	\$59.99
Controller	Radiomaster Zorro	\$119.99	1	\$119.99
Radio Receiver	Happymodel TCXO ExpressLRS 2.4GHz Nano Receiver	\$17.99	1	\$17.99
Video Transmitter	FPV VTX SpeedyBee TX800	\$27.99	1	\$27.99
Camera	Caddx Ratel 2 Micro Starlight 1200TVL	\$30.99	1	\$30.99
Goggles	EV800D FPV Goggles with DVR 5.8G 40CH	\$98.99	1	\$98.99
GPS Module	HGLRC M100 Mini GPS	\$15.99	1	\$15.99
Servo Transmitter	HOTRC DS-600 6CH 2.4Hz RC Transmitter and Receiver Remote Controller	\$29.96	1	\$29.96
Claw	Mechanical Robot Arm Claw/Gripper (Black Mechanical Claw with Servo)	\$32.99	1	\$32.99
			Parts Price	\$1,075.93
			Tax	\$67.25
			Final Cost	\$1,143.18

For our drone, we carefully selected each component to ensure optimal performance, stability, and efficiency during the competition. Below is an explanation of our choices and how they contribute to our drone's success in the challenge.

1. TBS Source One V5 5" Frame: We chose this frame for its durability, versatility, and affordability. The design provides ample space for components and ensures the drone's strength during flights and impacts, making it suitable for the challenge's requirements.
2. iFlight BLITZ Stack - F7 V1.1 FC + E55 4-in-1 55A ESC: This stack was selected for its advanced flight control features, efficient power management, and easy integration. The F7 flight controller offers precise and stable flight, while the 4-in-1 ESC ensures reliable power distribution to the motors.
3. iFlight XING-E Pro 2207 Motor (1800kv): These motors provide an excellent balance of power, efficiency, and responsiveness, allowing the drone to navigate the course with ease while maintaining the necessary speed and agility to complete the tasks.
4. FPV VTX SpeedyBee TX800: We chose this video transmitter for its adjustable power output and reliable performance, ensuring clear and stable video transmission during the challenge.
5. Mechanical Robot Arm Claw: This claw was the most efficient way of picking up our targets, which are the bean bags. It closes tightly to not let any of our targets slip away. It moves smoothly and quickly.
6. 3D Modeled Landing Gear: The gear was custom modeled to provide a flat landing surface that would allow our claw to reliably pick up all targets without the claw touching the ground.

Research of Rules

The use of drones for recreational and commercial purposes has expanded rapidly in recent years, leading to a growing need for regulations to ensure the safe and responsible operation of these unmanned aircraft systems (UAS). In the context of our UAV Drone Carnival Games project, our team strictly adhered to all applicable local, regional, and federal drone flight regulations. These regulations are in place to protect public safety, maintain privacy, and avoid interference with other aircraft.

To ensure that all team members operating the drone had a basic understanding of drone safety, regulations, and best practices, each member successfully completed the FAA's The Recreational UAS Safety Test (TRUST) exam. This certification provided a solid foundation for our team to operate the drone safely and responsibly throughout the project.

During the operation of our drone, we maintained a visual line of sight (VLOS) with the drone at all times, as required by the FAA for recreational drone pilots. This allowed us to maintain control over the drone and avoid potential collisions with other aircraft or obstacles. Maintaining VLOS also ensured that we were aware of our surroundings and could react quickly to any unexpected situations that might arise during flight.

We strictly adhered to altitude limits of 400 feet above ground level (AGL) and avoided flying in controlled airspace, near airports, or over groups of people, stadiums, or emergency response efforts. In addition to these restrictions, our team respected individual privacy by not flying over private properties without permission. This approach demonstrated our commitment to respecting the rights of others while operating the drone.

To ensure that our testing and practice flights were conducted in safe and permissible locations, we consulted the FAA's B4UFLY app, which provides information on restricted or sensitive areas. By using this resource, we were able to

avoid no-fly zones and ensure that our flights were compliant with all applicable regulations.

Weather conditions play a crucial role in the safe operation of drones, so our team operated the drone only in favorable conditions. We avoided flights during high winds, rain, or low visibility to ensure safety and minimize the risk of accidents. This consideration allowed us to maintain the integrity of our drone and reduce the likelihood of damage or injury resulting from adverse weather.

Before each flight, we conducted thorough preflight inspections of the drone to ensure all components were in proper working condition and the battery was fully charged. These checks allowed us to identify any potential issues before they could become more significant problems during flight and ensured that our drone was always ready for operation.

Our team's adherence to flight regulations throughout the UAV Drone Carnival Games project was a critical aspect of our commitment to safety and responsible drone operation. By following the rules set forth by the FAA and other relevant authorities, we were able to navigate the project successfully while maintaining compliance with all applicable regulations. As drones continue to grow in popularity and become an increasingly integral part of various industries, it is essential that operators prioritize safety and regulatory compliance to ensure the continued responsible use of this exciting technology.

Resources

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2. Gibson, J. (2020). FPV Flight Dynamics: Mastering Acro Mode on High-Performance Drones. Stone Blue Airlines.
3. Horvath, D. (2019). Drone Building Bible: A Step-By-Step Guide to Building Your Own Drone. CreateSpace Independent Publishing Platform.
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<https://www.modelaircraft.org/sites/default/files/100.pdf>.
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<https://www.fcc.gov/general/general-rules-unlicensed-wireless-equipment-operating-ism-bands>.

Work Log

Date	Task	Members Present	Comments	Time
2024-02-05	Project Kickoff and Team Formation	All members	Roles and responsibilities assigned. Discussed last year's successes and failures.	4 hours
2024-02-12	Reevaluate components	All members	Made a list of potential part efficiency need replacements for	4 hours
2024-02-19	Claw Design	Design team	Reassessed the efficiency of the old claw and brainstormed new iterations	6 hours
2024-02-26	Assembled new Components and Wiring	Assembly team	Put together the new claw attachment and tested in flight	4 hours
2024-03-04	Programming and Betaflight Setup	Programming team	Configured flight controller settings, PID tuning, and FPV system.	7 hours
2023-03-11	Carnival Game Simulation Testing	Testing team, Assembly team	Practice flight in the competition scenario	8 hours
2023-03-18	Sponsorship Outreach	All members	Individually reached out to companies for sponsorships.	6 hours

1-888125, 2-604857, 3-888145, 4-592326, 5-888146, 6-712494

Student Copyright Checklist

STUDENT COPYRIGHT CHECKLIST

(for students to complete and advisors to verify)

- 1) Does your solution to the competitive event integrate any music? ☐ YES ☒ NO

If NO, go to question 2.

If YES, is the music copyrighted? ☐ YES ☐ NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission (OR use royalty free/your own original music) and if permission is granted, include the permission in your documentation.

1B) Is the music royalty free, or did you create the music yourself? If YES, cite the royalty free music OR your original music properly in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any music into his/her competitive event solution.

I, Katelynn Robbins (chapter advisor), have checked my student's solution and confirm that the use of music is done so with proper permission and is cited correctly in the student's documentation.

- 2) Does your solution to the competitive event integrate any graphics? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is the graphic copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphic in your solution and included that permission (letter/form) in your documentation? If YES, move to question 3. If NO, ask for permission (OR use royalty free/your own original graphic) and if permission is granted, include the permission in your documentation.

2B) Is the graphic royalty free, or did you create your own graphic? If YES, cite the royalty free graphic OR your own original graphic properly in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any graphics into his/her competitive event solution.

I, Katelynn Robbins (chapter advisor), have checked my student's solution and confirm that the use of graphics is done so with proper permission and is cited correctly in the student's documentation.

- 3) Does your solution to the competitive event use another's thoughts or research? ☐ YES ☒ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? If YES, this is the end of the checklist.

If NO, properly cite the thoughts/research of others in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any thoughts/research of others into his/her competitive event solution.

I, K. Robbins (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation.